

Page 1 – Static Keyword (Interview Notes)

What is static?

static belongs to the class, not individual objects. One shared copy exists for the class.

Key Points

- One copy per class
- Loaded when class loads
- Shared by all objects
- Access using `ClassName.member`

Example

```
class Employee { static String company="Google"; int id; }
```

Interview Questions

- Why is `main()` static?
- Can constructors be static? (No)
- Can static methods be overridden? (No, only hidden)

30-Second Answer

static members belong to the class, are shared by all objects, and can be accessed without creating an object.

Page 2 – Static Variables & Methods

Static Methods

Belong to the class and can be called without creating an object.

Can Access

- ☐ Static variables
- ☐ Static methods
- ☐ Instance variables/methods directly

Why?

A static method has no object (no this reference).

Interview Questions

- Can an instance method call a static method? (Yes)
- Can a static method call an instance method? (Only using an object)

30-Second Answer

Instance methods have an object context, static methods do not.

Page 3 – Static Rules

Rules

- Access static members using the class name.
- Access modifiers still apply to static members.
- private static members cannot be accessed outside the class.

Common Mistake

Calling static members using an object is allowed but not recommended.

Rapid Fire

- Can static access instance? No.
- Can instance access static? Yes.

Page 4 – Static Nested Class

Definition

A class declared inside another class using the static keyword.

Why Use?

When the nested class doesn't need access to the outer object's state.

Interview Questions

- Can a top-level class be static? (No)
- Difference between static nested class and inner class?

30-Second Answer

Static nested classes don't require an outer object and directly access only outer static members.

Page 5 – Wrapper Classes

What are Wrapper Classes?

Wrapper classes convert primitive data types into objects.

Primitive	Wrapper
boolean	Boolean
byte	Byte
char	Character
short	Short
int	Integer
long	Long
float	Float
double	Double

Why Needed?

Collections, Generics and Streams work with objects, not primitives.

Autoboxing

```
Integer n = 10;
```

Unboxing

```
int x = n;
```

Interview Questions

- Why not ArrayList<int>?
- What is autoboxing?
- What is unboxing?

30-Second Answer

Wrapper classes are object forms of primitive types, enabling primitives to work with Java APIs requiring objects.